

# Tommaso Giacomello

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## EDUCATION

**Bocconi University – Milan, Italy**

**Sep 2026 - Present**

**PhD in Computer Science**

- Research: Transformer models for understanding gene regulation and synthetic biological data generation.
- Core courses: Computer Science, Modern Machine Learning, Bayesian Statistics, Statistical Physics, Stochastic Processes.
- Recipient of PhD scholarship from Fondo Italiano per la Scienza (FIS) for DECODE project.
- Supervised by Prof. Andrea Tangherloni.

**Bocconi University – Milan, Italy**

**Sep 2024 - Oct 2026**

**MSc in Artificial Intelligence**

**GPA: 28.1/30**

- Thesis: “*Gene Regulatory Network Inference from Enformer-based models*”.
- Core courses: Bioinformatics, Deep Learning, Reinforcement Learning, Natural Language Processing (NLP), Computer Vision and Software Engineering.

**Bocconi University – Milan, Italy**

**Sep 2021 - Jul 2024**

**BSc in Mathematical and Computing Sciences for Artificial Intelligence**

**Grade: 106/110**

- Thesis: “*Enriched Gene Regulatory Network algorithm applied to cancer cell dataset*”.
- Core courses: Algebra, Analysis, Physics, Statistics, Machine Learning, Optimization, Algorithms, Neuroscience.

## RESEARCH EXPERIENCE

**University of Oxford – Oxford, UK**

**Feb 2026 – May 2026**

**Visiting Researcher – Weatherall Institute of Molecular Medicine**

- Applied pre-trained transformer-based models (Enformer, AlphaGenome) to infer cell-type specific Gene Regulatory Networks from multi-omics epigenomic data (RNA-seq, ATAC-seq, TF ChIP-seq).
- Worked with large-scale datasets and HPC infrastructure (SLURM, GPU-accelerated training).
- Supervised by Prof. Jim Hughes and Simone Riva.

**Bocconi University – Milan, Italy**

**Sep 2024 – Present**

**Assistant Researcher – Department of Computing Sciences**

- Developed machine learning methods for Gene Regulatory Network inference using scRNA-seq and scATAC-seq data.
- Extended the CellOracle framework with enrichment-based methods to improve biological interpretability.
- Awarded three research contracts under the ERC-funded MicroC project.
- Supervised by Prof. Francesca Buffa.

## PUBLICATIONS

**Giacomello et al. (2026): “pysigscore: gene signatures scoring across bulk and single-cell transcriptomics” (Under Review)**

- Developed Python package (pysigscore) for computing signature scores in bulk and scRNA-seq data.
- Extended functionality of the original R library (alebarberis/sigcores) and deployed as a PyPI package.

**Giacomello et al. (2025): “Enrichment Algorithms for Gene Regulatory Network Reconstruction” CIBB 2025**

## PROJECTS

**Investigating Public Concerns About AI**

**NLP Group Project**

- Analysed public concerns about AI by building a 20k+ tweet and 500+ article dataset.
- Fine-tuned Twitter-RoBERTa and DistilBERT (~70% F1-score on negative sentiment classification).
- Applied topic modelling (NMF, LDA) to extract structured themes in media framing.

**TP53 Mutation Prediction**

**AI Lab Group Project**

- Built a pipeline for preprocessing and batch correction of CCLE (Cancer Cell Line Encyclopedia) datasets.
- Predicted TP53 mutation status using Random Forest and XGBoost (~81% F1-score), optimised using Optuna library.
- Project work later used by Prof. Salwa Lin in a research grant proposal.

**Deep Learning for Sign Language Prediction**

**Deep Learning Project**

- Processed World Level American Sign Language dataset (WLASL), 21k+ videos and 2000+ signs.
- Achieved 60% F1-score on 20 sign classification using 2D-CNN on hand landmarks.

## SKILLS

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|------------------------|-------------------------------------------------------------------------------------------------|
| <b>Languages:</b>      | Italian (Native), English (C1 Fluent, IELTS 7.5), Spanish (B1 Business, University Exam 27/30). |
| <b>Programming:</b>    | Python, C, R.                                                                                   |
| <b>AI Libraries:</b>   | NumPy, Pandas, PyTorch, XGBoost, LangChain, Scikit-Learn, Optuna, Keras, Captum, Hugging Face.  |
| <b>Bioinformatics:</b> | Scanpy, CellOracle, deepTools.                                                                  |
| <b>Tools:</b>          | Linux shell, Git, HPC (SLURM), GPU computing.                                                   |

## PRIVACY

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I authorize the use of my personal data according to GDPR 679/2016.